

MEMSYST STAGE 11 DEVELOPMENT MASTER PROMPT

Events, Conferences, Meetings & Attendance Management Platform (ECMAP)

You are a Senior Event Management Architect, Membership Engagement Architect, Enterprise SaaS Architect, Digital Governance Architect, and Principal Engineer responsible for building Stage 11 of the MemSyst ecosystem.

Stages 1–10 have already established:

Stage 1

Business Operations Platform

Stage 2

Identity & Access Platform

Stage 3

Tenant Management Platform

Stage 4

Membership & Apprentice Platform

Stage 5

Financial Infrastructure Platform

Stage 6

Communication & Engagement Platform

Stage 7

Digital Identity & Credential Platform

Stage 8

Marketplace, Business Directory & Commerce Platform

Stage 9

Training, Learning, Apprenticeship & Skills Development Platform

Stage 10

Support Hub, Opportunities, Grants, Loans & Member Services Platform

Stage 11 introduces the event, meeting, conference, convention, and attendance infrastructure of the MemSyst ecosystem.

This stage enables organizations to manage everything from local branch meetings to national conferences while integrating attendance, registration, payments, communications, credentials, and governance.

STAGE 11 OBJECTIVE

Build the:

Events, Conferences, Meetings & Attendance Management Platform (ECMAP)

Purpose:

Enable organizations to:

Create Events

Manage Conferences

Manage Meetings

Register Participants

Track Attendance

Issue Event Credentials

Collect Event Fees

Generate Event Reports

Manage Event Communications

while integrating with all previous platform capabilities.

CRITICAL EVENT MANAGEMENT PRINCIPLES

Principle 1

Events Are Tenant-Specific

Every event belongs to a tenant.

Examples:

GHABA Annual Congress

Regional Conference

Branch Meeting

Executive Retreat

Training Workshop

Events are isolated within their tenant environment.

Principle 2

Event Types Must Be Configurable

No hardcoded event structures.

Each tenant can create:

Conferences

Meetings

Workshops

Seminars

Trade Shows

Exhibitions

Award Ceremonies

Training Events

Custom Event Types

Principle 3

Attendance Must Be Verifiable

Attendance records must be:

Auditable

Trackable

Verifiable

and linked to:

Member Profiles

Credentials

Training Records

where applicable.

Principle 4

Event History Is Permanent

Events, registrations, and attendance records must never be deleted.

Historical participation forms part of member engagement history.

STAGE 11 DELIVERABLES

Build:

Event Management Engine
Conference Management Engine
Meeting Management Engine
Registration Engine
Attendance Engine
Ticketing Engine
Event Credential Engine
Event Analytics Engine

MODULE 1

EVENT MANAGEMENT ENGINE

Purpose:

Create and manage events.

Event Record

Event ID
Tenant ID
Event Type
Title
Description
Venue
Start Date
End Date
Status

Status

Draft
Published
Open Registration
Closed Registration
Ongoing
Completed

Cancelled

Archived

Features

Create Event

Edit Event

Publish Event

Cancel Event

Archive Event

Clone Event

MODULE 2

EVENT CATEGORIES ENGINE

Purpose:

Support tenant-specific event structures.

Examples

National Congress

Regional Meeting

Branch Meeting

Training Workshop

Seminar

Exhibition

Award Event

Requirements

Categories must be:

Tenant Configurable

Version Controlled

MODULE 3

EVENT REGISTRATION ENGINE

Purpose:

Manage participant registration.

Registration Sources

Member Self Registration

Executive Registration

Administrator Registration

Invitation Registration

Registration Record

Registration ID

Participant

Event

Registration Date

Status

Payment Status

Attendance Status

Registration Status

Pending

Confirmed

Waitlisted

Cancelled

Attended

No Show

MODULE 4

EVENT TICKETING ENGINE

Purpose:

Manage event access.

Ticket Types

Free Ticket
Paid Ticket
VIP Ticket
Delegate Ticket
Executive Ticket
Custom Ticket

Ticket Record

Ticket ID
Event
Holder
Ticket Type
QR Code
Status

Ticket Status

Issued
Checked In
Cancelled
Expired

MODULE 5

EVENT PAYMENT ENGINE

Purpose:

Manage event-related payments.

Payment Types

Registration Fee
Conference Fee
Workshop Fee
Exhibition Fee
Certification Fee

Integration

Must connect directly with:

Stage 5 Wallets

Billing Engine

Revenue Distribution Engine

Receipts Engine

Features

Fee Collection

Receipt Generation

Refund Support

Financial Reporting

MODULE 6

ATTENDANCE MANAGEMENT ENGINE

Purpose:

Track event participation.

Attendance Methods

QR Code Check-In

Manual Check-In

Executive Verification

Digital Attendance

Attendance Record

Participant

Event

Check-In Time

Check-Out Time

Attendance Status

Attendance Status

Present
Absent
Late
Excused

MODULE 7

QR CHECK-IN ENGINE

Purpose:

Provide fast event access and attendance validation.

QR Sources

Member ID Card
Event Ticket
Event Credential

Features

Scan
Validate
Check-In
Prevent Duplicate Entry
Attendance Logging

MODULE 8

EVENT CREDENTIAL ENGINE

Purpose:

Issue event participation credentials.

Credential Types

Attendance Certificate
Conference Certificate
Speaker Certificate

Volunteer Certificate
Organizer Certificate

Integration

Must generate credentials through:

Stage 7 Credential Platform

Triggers

Attendance Confirmed
Event Completed
Organizer Approval

MODULE 9

SPEAKER & FACILITATOR MANAGEMENT

Purpose:

Manage event presenters.

Record

Speaker ID
Name
Organization
Biography
Session Assignment
Status

Features

Assign Speaker
Manage Sessions
Publish Profiles
Track Participation

MODULE 10

SESSION MANAGEMENT ENGINE

Purpose:

Manage event programs and schedules.

Session Record

Session ID

Event

Title

Speaker

Venue

Date

Time

Capacity

Features

Create Session

Assign Speaker

Manage Capacity

Track Attendance

MODULE 11

INVITATION MANAGEMENT ENGINE

Purpose:

Manage invitations.

Invitation Types

Member Invitation

Executive Invitation

Guest Invitation

Speaker Invitation

Sponsor Invitation

Features

Send Invitation
Track Responses
Manage RSVP
Reminder Notifications

MODULE 12

EVENT COMMUNICATION INTEGRATION

Purpose:

Connect event management with Stage 6.

Notifications

Registration Confirmed
Payment Received
Event Reminder
Schedule Update
Attendance Confirmed
Certificate Available

Channels

Email
SMS
Push
In-App

MODULE 13

EVENT ANALYTICS ENGINE

Purpose:

Provide event insights.

Metrics

Registrations
Attendance Rate
Revenue
Ticket Sales
Session Participation
Speaker Participation

Scope

Platform
Tenant
Region
Branch
Event

MODULE 14

EVENT REPORTING ENGINE

Purpose:

Generate event reports.

Reports

Registration Report
Attendance Report
Financial Report
Speaker Report
Session Report
Certificate Report

Export Formats

PDF
Excel
CSV

MODULE 15

EVENT AUDIT ENGINE

Purpose:

Provide accountability.

Audit Events

Event Created

Event Updated

Registration Approved

Ticket Issued

Attendance Recorded

Certificate Issued

Audit Record

Actor

Action

Record

Date

Time

Before

After

MODULE 16

EXECUTIVE & GOVERNANCE EVENTS

Purpose:

Support governance-related meetings.

Examples

Annual General Meeting (AGM)

National Congress

Executive Council Meeting

Regional Meeting

Branch Meeting
Election Meeting

Additional Features

Restricted Attendance
Invitation Only Access
Executive Verification
Governance Reporting

MODULE 17

EVENT SETTINGS MANAGEMENT

Purpose:

Tenant-level event configuration.

Settings

Event Types
Registration Rules
Attendance Rules
Ticket Rules
Credential Rules
Fee Rules

Requirements

Everything configurable.

No hardcoded event workflows.

DATABASE DESIGN REQUIREMENTS

Before implementation:

Design complete entities for:

Events
Event Categories

Registrations
Tickets
Attendance
Sessions
Speakers
Invitations
Certificates
Event Audits
Event Settings

Define:

Relationships
Indexes
Validation Rules
Capacity Rules
Attendance Rules
Lifecycle States

SECURITY REQUIREMENTS

Integrate with Stage 2.

Enforce:

RBAC
Tenant Isolation
Event Permissions
Attendance Permissions
Financial Permissions
Audit Logging

FUTURE READINESS REQUIREMENTS

Stage 11 must prepare for:

Virtual Events
Hybrid Events
Live Streaming
Webinars
Online Ticket Sales
Event Sponsorship Management
Voting & Elections

Conference Mobile Apps

AI Event Assistants

Do not build them yet.

Design all required relationships now.

ARCHITECTURE REQUIREMENTS

Use:

Event-Driven Architecture

Tenant-Aware Services

Configuration-Driven Rules

QR-Based Validation

Audit-First Design

Avoid:

Hardcoded Event Types

Manual Attendance Spreadsheets

Untracked Registrations

Deleting Attendance Records

All event participation must be traceable and historically preserved.

SUCCESS CRITERIA

Stage 11 is complete only when:

- ✓ Events can be created and managed
- ✓ Conferences and meetings can be organized
- ✓ Registrations are operational
- ✓ Ticketing is operational
- ✓ QR check-in is operational
- ✓ Attendance tracking is operational
- ✓ Event payments integrate with Stage 5
- ✓ Event communications integrate with Stage 6
- ✓ Event credentials integrate with Stage 7

- ✓ Session and speaker management exist
- ✓ Event analytics exist
- ✓ Event reporting exists
- ✓ Event audit history exists
- ✓ Governance-related meetings are supported
- ✓ Platform is fully prepared for Stage 12 (Governance, Elections, Voting & Decision Management Platform)

Build this stage as the event and participation infrastructure of the MemSyst ecosystem, enabling organizations to manage everything from branch meetings to national congresses while maintaining complete accountability, engagement tracking, and integration with the broader MemSyst platform.